

ECE321

Electronic circuits

Lecture (06)- Op. Amp_Offset

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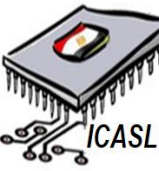


- ☐ DC imperfection
- ☐ Input Bias & Input Bias Offset
- ☐ Input Offset Voltage
- ☐ Exercise

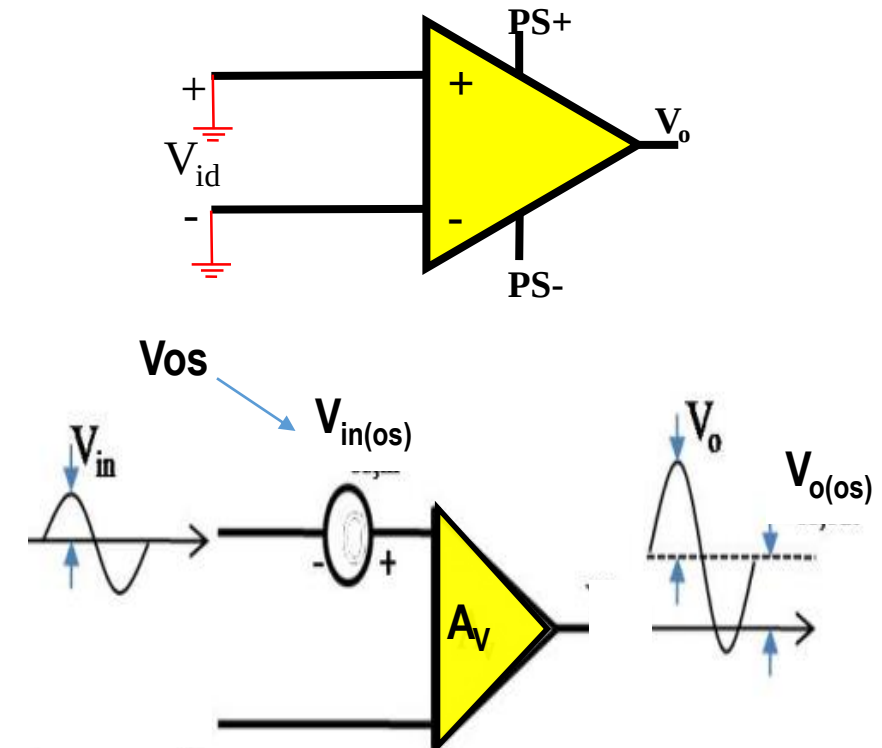
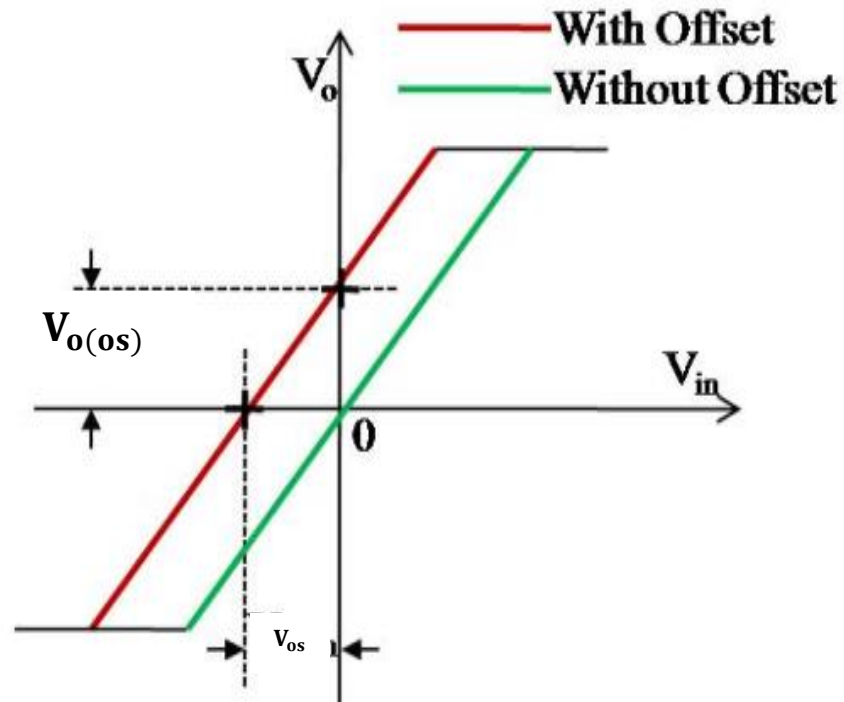


- ☐ **DC imperfection**
- ☐ **Input Bias & Input Bias Offset**
- ☐ **Input Offset Voltage**
- ☐ **Exercise**

DC Imperfection (1/3)



- ❑ Because op amps are **direct-coupled** devices with large gains at dc, they are **prone** to dc problems.
- ❑ To **understand** this problem consider the following **conceptual** experiment:
 - ❑ If the **two input terminals** of the op amp are tied together and connected to **ground**, it will be found that despite the fact that $V_{id} = 0$, a **finite dc voltage** exists at the output ($V_{o(os)}$)
- ❑ In fact, if the op amp has a **high dc gain**, the output will be at either the **positive** or **negative** saturation level
- ❑ The op-amp output can be brought back to its ideal value of 0 V by connecting a dc voltage source of appropriate polarity and magnitude between the two input terminals of the op amp.

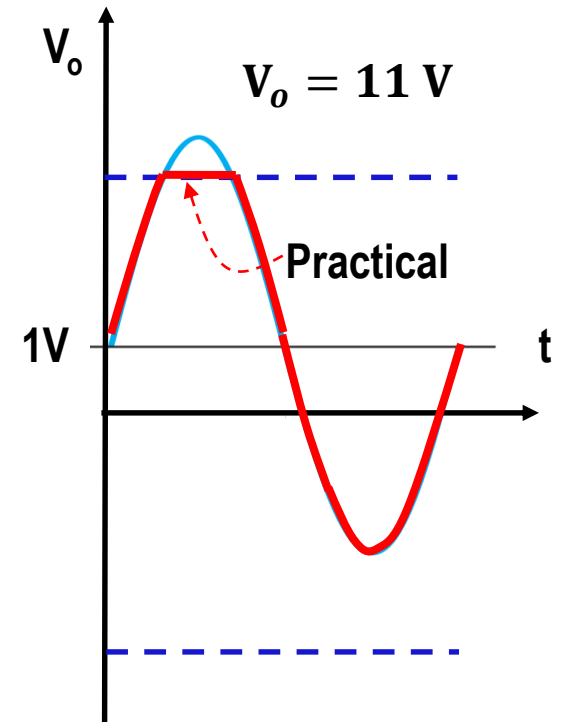


- $$V_{os} = \alpha (T - T_o)$$

T: Temperature @ calculating V_{OS}

- ☐ Reduce output sewing
- ☐ Distorted the output voltage
- ☐ Increase Total Harmonic Distortion (THD)
- ☐ Reduce Dynamic Range (DR)

$$\mathbf{V_o} = (\mathbf{V_{os}} + \mathbf{V_i})^* \mathbf{A_v}$$



DC Imperfection (3/3)



The important non-ideal properties of the op am

- Input Bias " I_B " & input Bias Offset " I_{os} "
- Input offset voltage " V_{os} "

***We need to understand
those offsets and how they
were solved.***

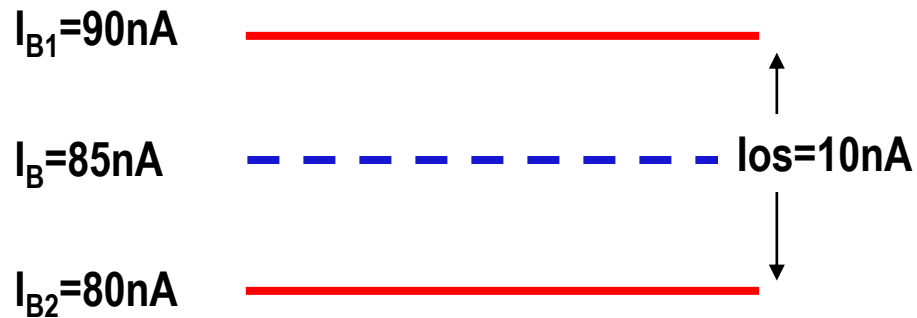


- ☐ DC imperfection
- ☐ **Input Bias & Input Bias Offset**
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Input Bias & Input Bias Offset (1/3)

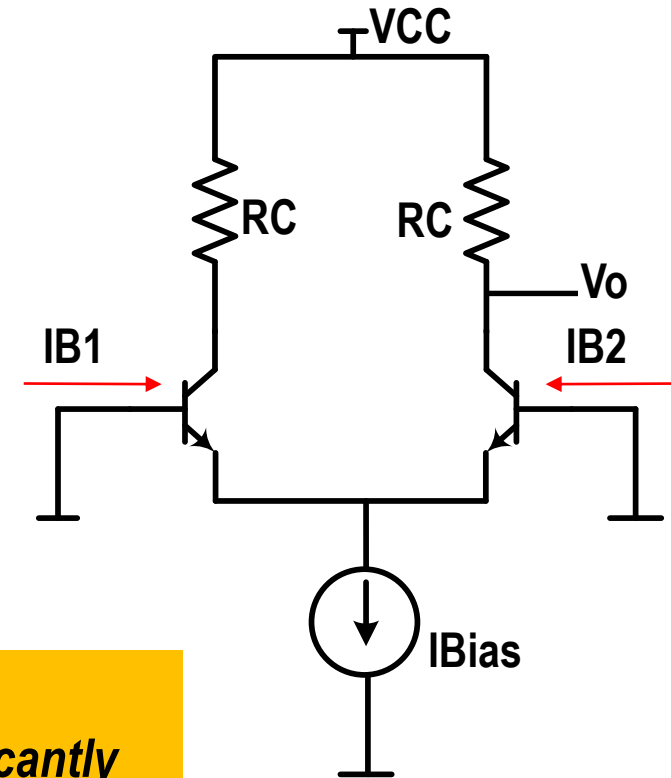
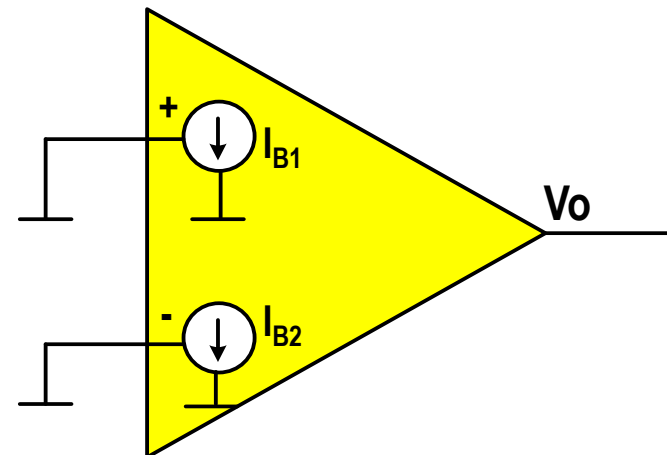


- ❑ In order for the op amp to operate, its two input terminals have to be supplied with dc currents, termed the input bias currents.
- ❑ These two currents are represented by two current sources, I_{B1} and I_{B2} , connected to the two input terminals.
- ❑ It should be emphasized that the input bias currents are independent of the fact that a real op amp has finite (though large) input resistance.
- ❑ The op-amp manufacturer usually specifies the average value of I_{B1} and I_{B2} as well as their expected difference.



$$I_B = \frac{I_{B1} + I_{B2}}{2} \quad \text{input bias current}$$

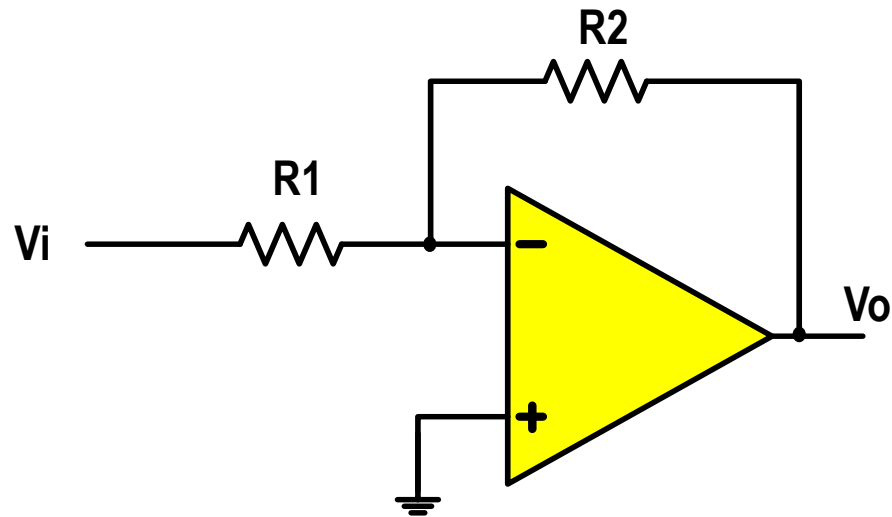
$$I_{OS} = |I_{B1} - I_{B2}| \quad \text{input offset current}$$



$I_{os} \lll I_B$
**BJT and JFET would effect significantly
with I_B and I_{os}**

Input Bias & input Bias Offset (2/3)

- Study effect of the input bias current and input bias offset current on the inverting amplifier.



Ans.

(A) Where is the Problem

Set $V_i=0$ to study the offset

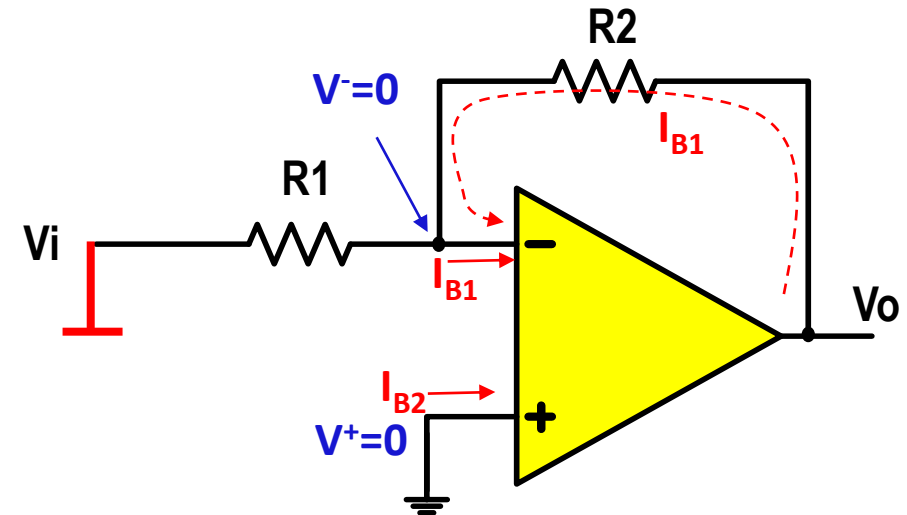
$V^+=V^-=0$

$I^- = I_{B1}$ and $I^+ = I_{B2}$



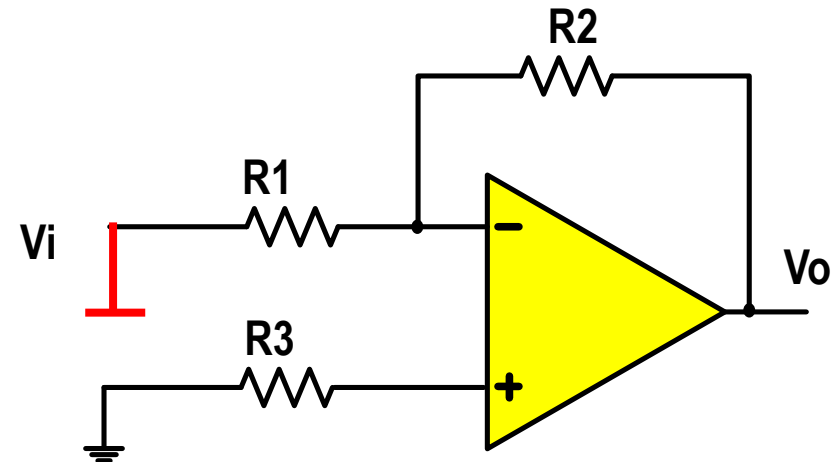
$$V_o = I_{B1} * R_2$$

Although the $V_i=0$, V_o has finite value ($V_o = I_{B1} * R_2$)

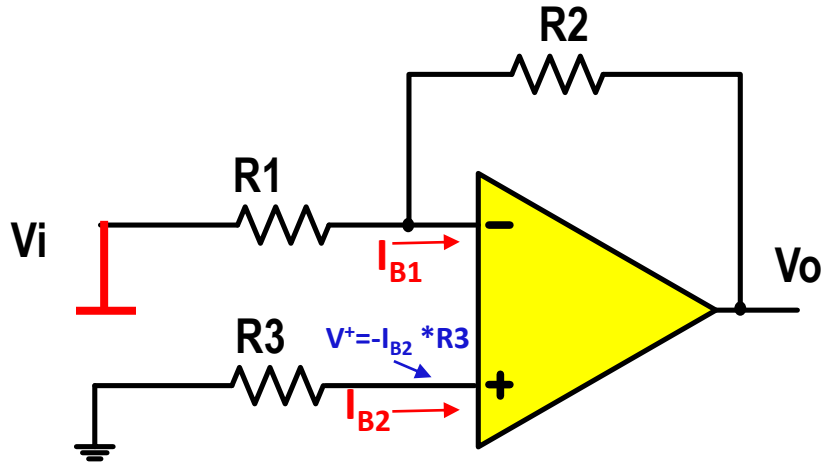


(B) Method to reduce this imperfection

By adding a resistor at V^+ as shown. Then calculate V_o again !



Input Bias & input Bias Offset (3/3)



Set $V_i=0$ to study the offset

$$V^+ = V^- = V^+ = -I_{B2} * R_3$$

$$I^- = I_{B1} \quad \text{and} \quad I^+ = I_{B2}$$

To get V_o , by taking a node @ V^-

$$I_{B1} + \frac{V^-}{R_1} + \frac{V^- - V_o}{R_2} = 0$$

$$I_{B1} + V^- \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = \frac{V_o}{R_2}$$

$$V_o = I_{B1} R_2 + V^- \left(\frac{R_2}{R_1} + 1 \right)$$

$$V_o = I_{B1} R_2 - I_{B2} R_3 \left(\frac{R_2}{R_1} + 1 \right)$$

$$V_o = I_{B1} R_2 - I_{B2} R_3 \frac{R_2}{R_1} - I_{B2} R_3$$

$$V_o = I_{B12} \left(R_2 - R_3 \frac{R_2}{R_1} - R_3 \right)$$

$$V_o = I_{B12} \left(R_2 - R_3 \frac{R_2}{R_1} - R_3 \right) = 0$$

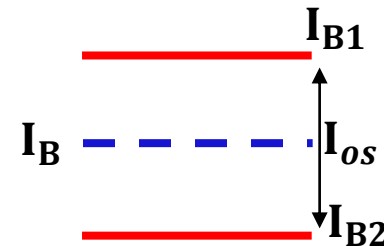
$$I_{B12} \neq 0 \quad \text{But} \quad \left(R_2 - R_3 \frac{R_2}{R_1} - R_3 \right) = 0 \Rightarrow R_3 = \frac{R_1 * R_2}{R_1 + R_2}$$



Return back and substitute by R_3 to equation (1) to make sure that V_o is reduced or not

$$V_o = I_{B1} R_2 - I_{B2} * \left(\frac{R_1 * R_2}{R_1 + R_2} \right) * \left(\frac{R_2}{R_1} + 1 \right)$$

Remember



$$I_{B1} = I_B + \frac{I_{os}}{2}$$

$$I_{B2} = I_B - \frac{I_{os}}{2}$$

$$V_o = I_{os} * R_2$$



Offset is reduced since $I_{os} \ll I_{B1}$



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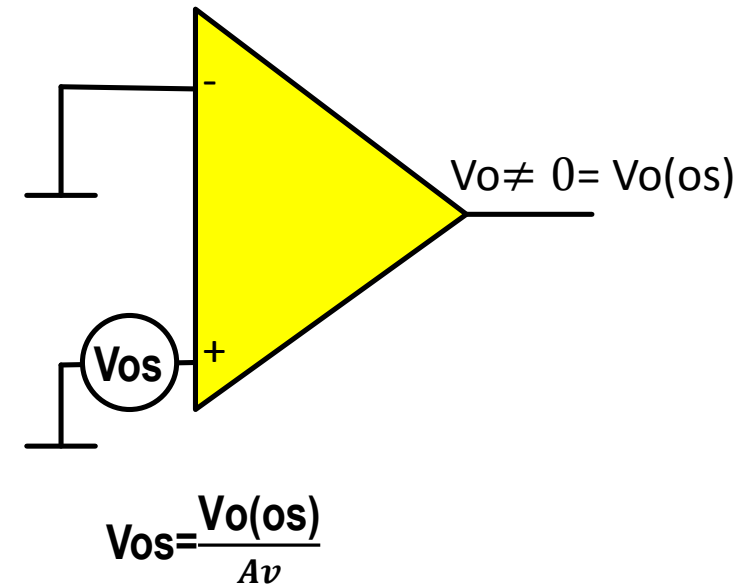
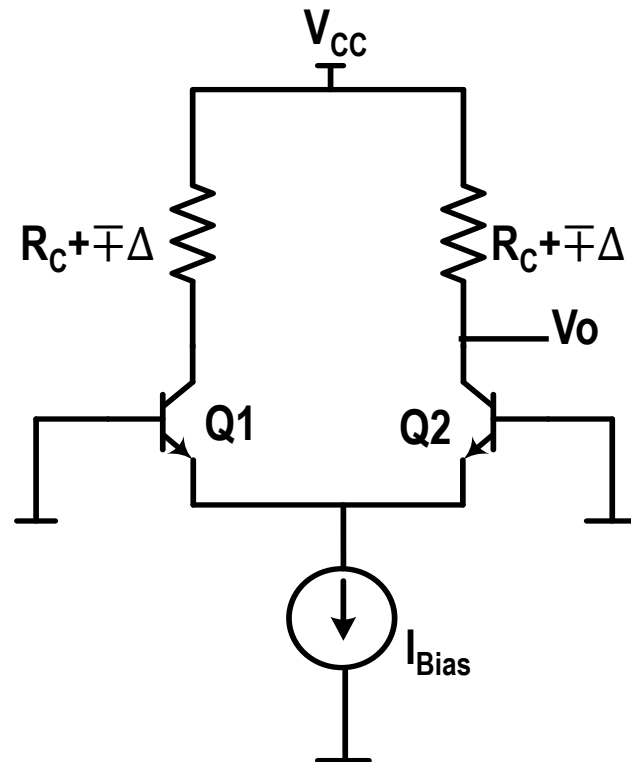
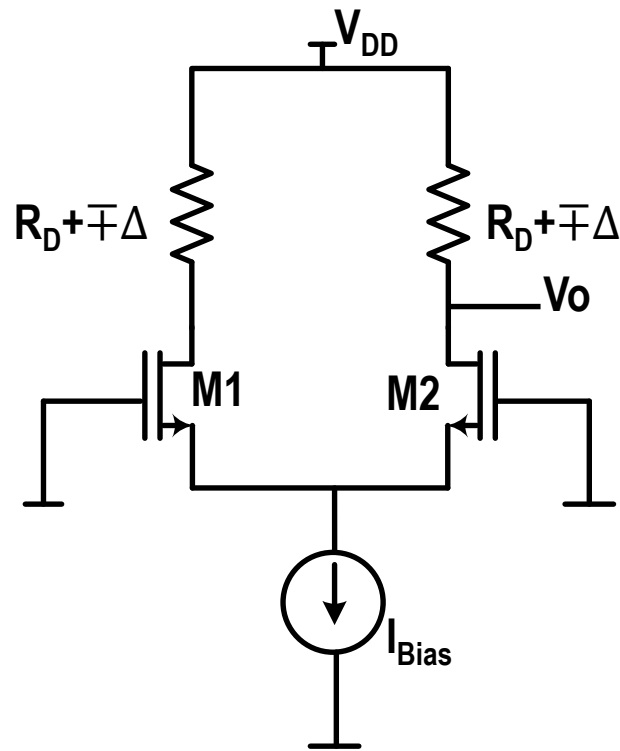
Input Offset Voltage (1/3)



❑ In this case, the offset is due mismatching between transistors and even resistive loads.

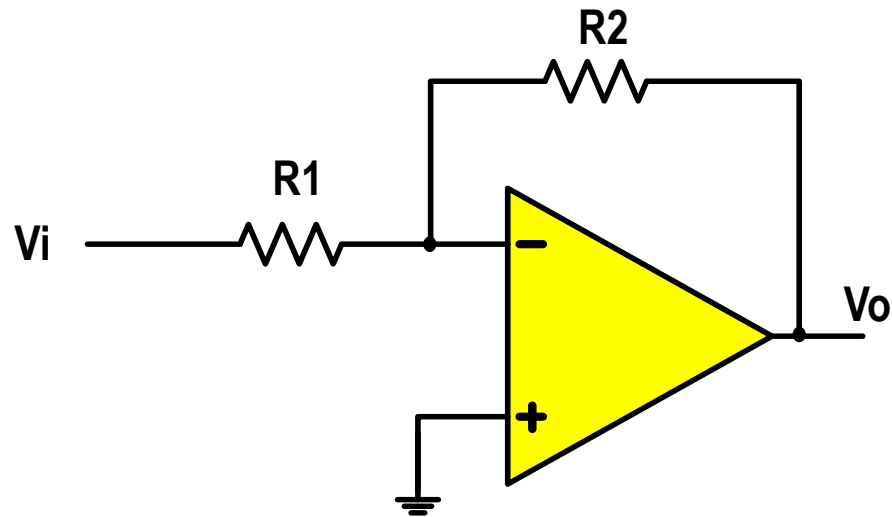
❑ $Q_1 \neq Q_2$ and $M_1 \neq M_2$

❑ $R_{C1} \neq R_{C2}$ and $R_{D1} \neq R_{D2}$



Input Offset Voltage (2/3)

Study effect of the offset voltage on the inverting amplifier.

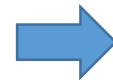


Ans.

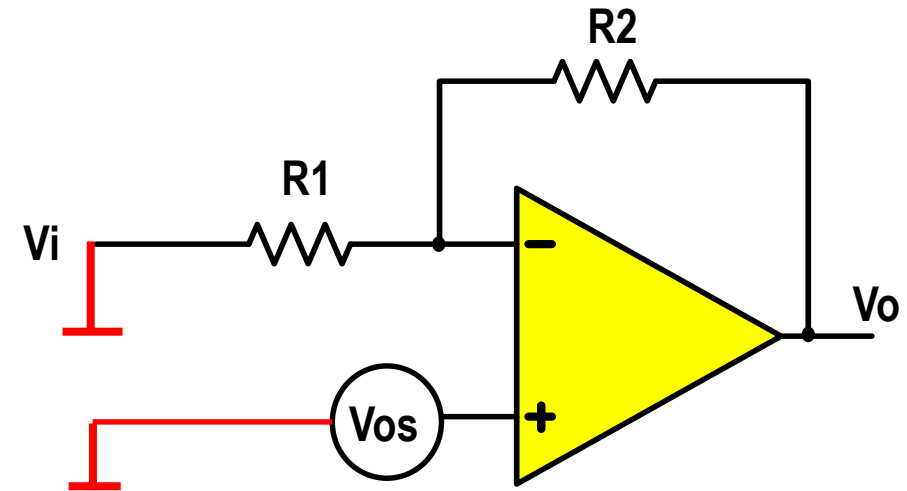
(A) Where is the Problem

Set $V_i=0$ to study the offset

$V^+=V^- \neq 0$ due to mismatching



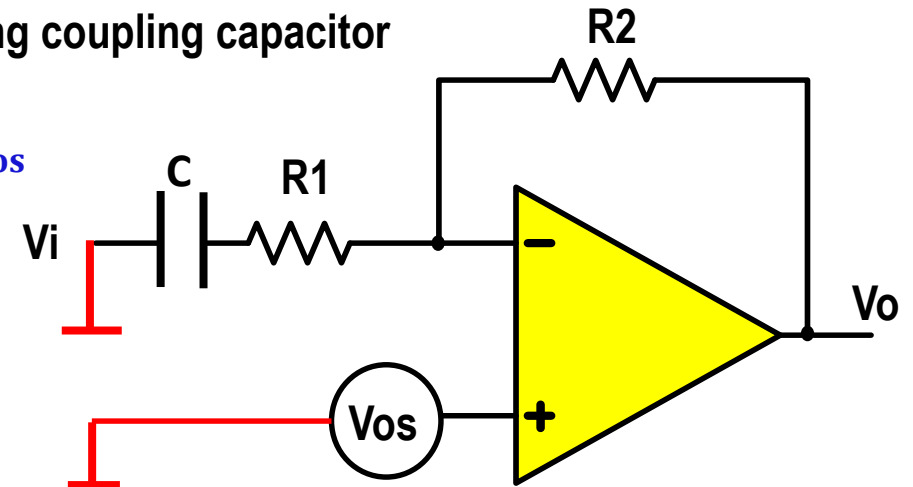
$$V_o = V_{os} \left(\frac{R_2}{R_1} + 1 \right)$$



(B) Method to reduce this imperfection

B.1 Using coupling capacitor

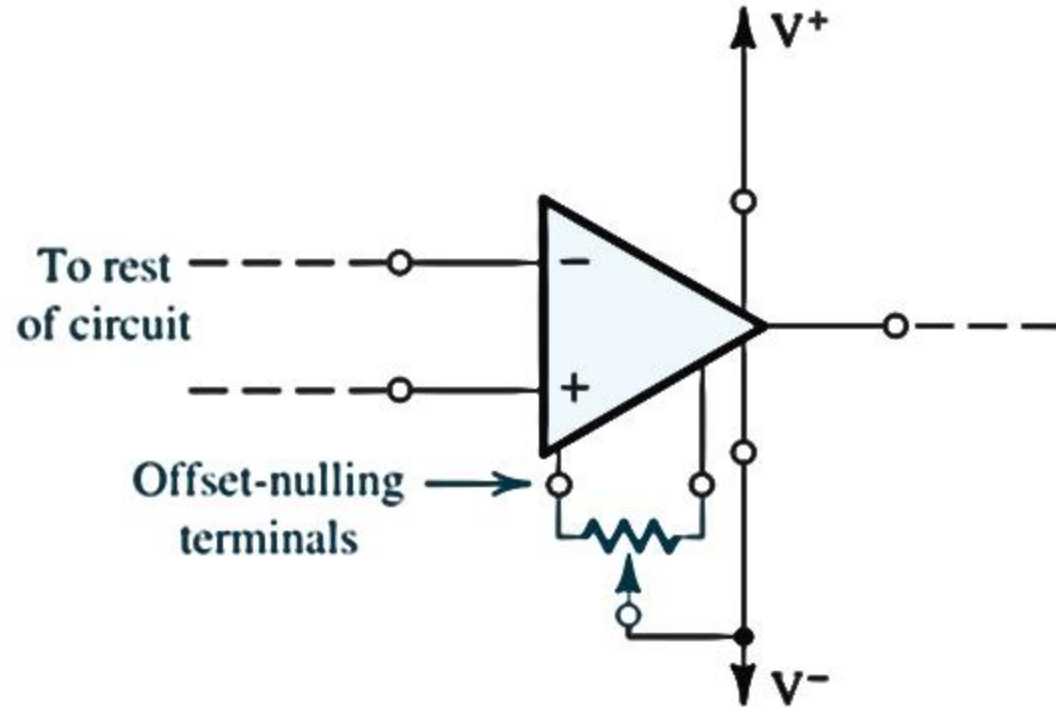
$$V_o = V_{os}$$



Input Offset Voltage (3/3)



B.2 Using a nulling terminals for LM741 [1,5]



The output dc offset voltage of an op amp can be trimmed to zero by connecting a potentiometer to the two offset-nulling terminals. The wiper of the potentiometer is connected to the negative supply of the op am.



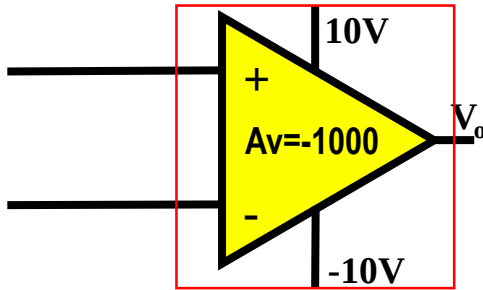
- ☐ DC imperfection
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Exercise



- Assume an inverting amplifier with close loop gain $A_v = -1000$, input offset voltage V_{os} of 3 mV, and the output saturation levels $= \pm 10$ V.
- What is the approximated peak sin wave input signal that can be applied without clipping?

Ans.

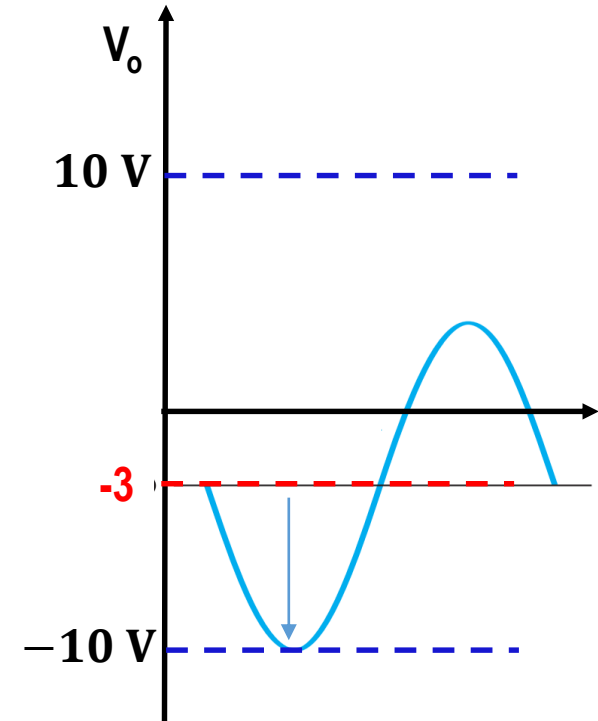


$$V_{os} = V_{in}(os) = 3\text{mV}$$

$$V_o = V_i \cdot A_v$$

$$V_o(os) = V_{in}(os) \cdot A_v$$

$$V_o(os) = 3\text{m} \cdot -1000 = -3\text{ V}$$



$$V_{o,Peak} = -7\text{ V}$$

$$V_{in,Peak} = V_{o,Peak} / A_v \\ = -7 / (-1000) = 7\text{mV}$$

Thanks!